Advanced Health Informatics

Engineering Design

2024

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Identification of Problem

The grand engineering challenge given, *Advance Health Informatics,* is a broad spectrum of topics, ranging from collecting patients' medical records to managing and analyzing the data. This challenge has a large span of topics we could cover, but we strived to choose one useful to a large portion of the population.

One of the main problems regarding Health Informatics is mental health. It is an often overlooked aspect of health, yet a highly important one. 1 in 5 U.S. adults experience mental illnesses each year, with 25% of them being serious. Suicide was the eleventh leading cause of death overall in the United States, claiming the lives of over 48,100 people. While self-diagnosis is an option, patients often end up making incorrect assumptions. This can be dangerous, especially with mental health, leading to distress and a list of solutions that won't work.

Even if someone goes to a medical professional to accept help, there is always that discomfort of speaking to someone so openly about your feelings. Oftentimes, people don't share the full truth when talking to a person face-to-face, and there usually isn't enough time for the doctor to fully understand the patient's problem.

This is also partially true for the long list of yes or no questions that you are given in the doctor's office lobby. Most people answer these questions by choosing the more positive of the choices because, even though they are not looking at a person, they would feel uncomfortable to answer anything except what they think *should* be the answer. The problem here is that the questions are too straightforward and blunt

This all contributes to the need for a better way to give patients better, accessible, accurate solutions to help them with mental illnesses and save doctors' time so they can jump right to the questions that only a professional could dissect and properly diagnose the patient.

Information Gathering

Health informatics can play a significant role in addressing depression by facilitating early detection and intervention, personalized treatment plans, remote monitoring and telemedicine, data-driven research, and patient engagement. Advanced analytics on patient data can help identify signs of depression earlier, which allows for early support, and personalized treatment plans can be made based on this individual patient data. Remote monitoring and telemedicine enable continuous support without the need for frequent in-person visits, saving doctors a lot of time. Large-scale data analysis can contribute to understanding the complex factors contributing to depression, informing the development of new treatments and preventive measures. The more data you have, the more predictions one can make.

For example, the suicide rates throughout the years have been increasing alongside the global population demographic. Through health informatics, information about suicidal thoughts and symptoms of depression have been recorded in a series of datasets. These datasets can be used to train Machine Learning models to predict suicide, and used to educate doctors on the symptoms of suicide. With this newfound information, doctors and Machine Learning models can accurately predict if a patient has depression and suicidal thoughts, and provide them with the help they need.

In the 1950s, the introduction of computers in healthcare began the integration of technology into medical records management, laying the groundwork for health informatics. The 1960s and 1970s saw the digitization of patient information, while the 1980s marked the beginning of Electronic Health Records (EHRs). The 1990s saw the rise of the internet and increased efforts to connect healthcare systems. The early 2000s witnessed widespread EHR adoption. The use of data analytics and machine learning in health informatics has led to improved patient care, reduced errors, and better healthcare quality.

Health informatics tools can empower patients by providing access to information and resources, enhancing patient engagement and mental health. The ongoing development of health informatics promises continuous advancements in healthcare efficiency and accuracy.

Through tests and other processes, we can gain access to information about the patient, including their condition. For example, doctors can assign the questionnaire to the patient, which will include a series of multiple choice questions that allow the doctor to evaluate the condition of the patient. As a result, it potentially acts as a dataset for doctors, and they can decide to diagnose whether the patient should be diagnosed with a specific condition or not. Moreover, this dataset can aid in looking at what symptoms or scenarios are the most problematic with the patient, and the doctor can look at what they should spend time on.

Possible Solutions

Finding solutions for an issue that is not tangible by touch was rather challenging, but through multiple brainstorming sessions, we came to three main possible solutions.

One possible solution to this problem is a watch that tracks the physical effects on the body that is typically caused by depressive disorders. This device would need to be worn at all times to produce accurate results and must have the technology capable of tracking one's heartbeat, sleep, increase or decrease in food intake, times of movement or lack of movement, and so on. These irregularities are symptoms of most depressive disorders, according to our Information Gathering. However, there are many other symptoms that must also be taken into account when considering Mental Health. This includes how one feels in certain scenarios, which could even be a valid cause of an irregularity. The device must also be worn at all times to collect a sufficient amount of data to analyze practical “problems” in the patient’s day to day behavior. Most people do not wear watches twenty-four hours a day, and this may cause the analysis to have a large margin of error. Moreover, the production costs of such an accurate product would be rather expensive, hence causing the cost of the device to also be unaffordable to many.

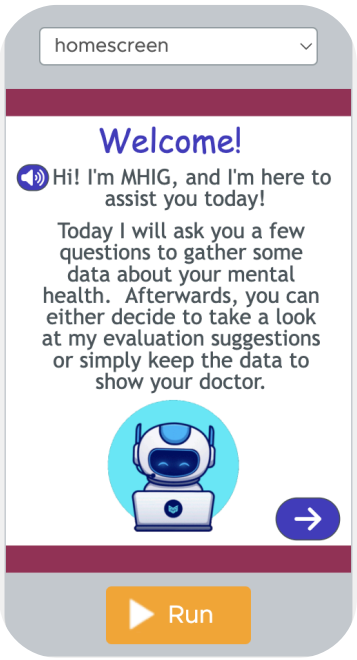
Another possible solution to this problem would be Brain Computerized Interface technology detecting changes in the brain during regular doctor appointments. However, there are most likely many practical difficulties and issues with this technology as well. First of all, the machine learning system would have to be fed a monstrous amount of data to produce an accurate analysis. Not only is the environment for building such a prototype rather challenging, but the product must be tested before being produced as our final solution. Testing would be difficult due to ethical concerns that surround the topic. Production costs would be rather high as well, but the number of products would be relatively lower due to only hospitals owning this technology.

Moreover, in a society where Mental Health is devalued, many would feel that it is unnecessary to perform such big procedures for an aspect that could be evaluated much easier. Though this solution had its advantages, we felt as though there could be a better solution.

The third and selected solution to the challenge is to create an application that can collect patient data discretely while assuring that the patient is not uncomfortable. This data can then be used to make an analysis that can aid doctors in helping diagnose and identify depressive disorders. We can present our solution by researching mental disorders, developing questions that are easy to answer, and displaying them in an understandable and accessible program for the patients. Through the app, doctors can access information about their feelings and irregularities in health related to mental health disorders, which are collected by many multiple-choice questions. Then the doctors can analyze the symptoms provided by the patients and make a decision, or a diagnosis of the patient. This solution is cost effective and flexible as well since people could take the program's test from home or in the hospital, however the health care providers feel would be best for accurate gathering of the symptoms. The solution fits all of our criteria and constraints as well, making it the best solution out of our brainstorm.

Visual Representation of Prototype/Model

Prototype 1



Prototype 2

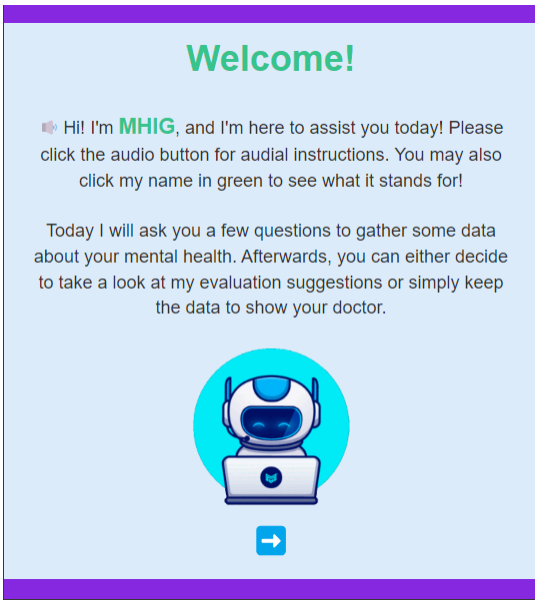
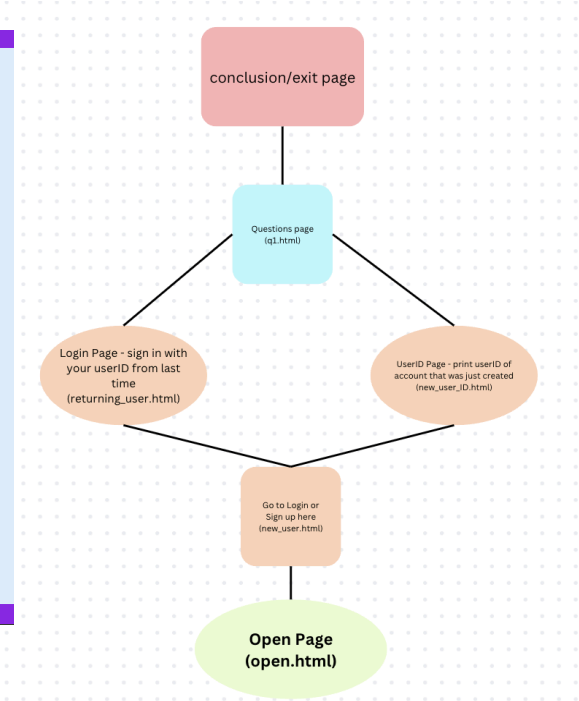
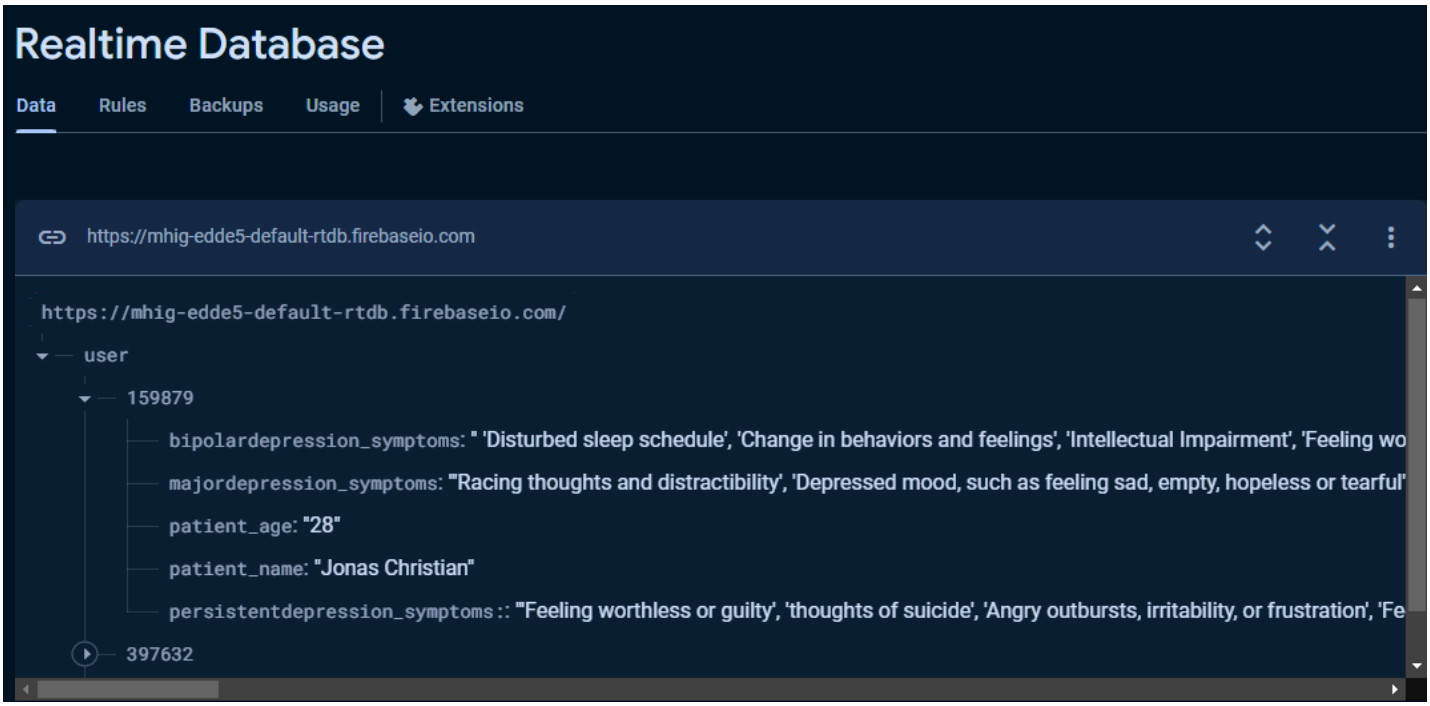


Diagram of Page Flow





Summary of Iteration Process

Define the Problem: Mental health is disregarded during regular doctor check ups, causing drastic outcomes. How can technology be integrated into engineering practices to enhance the processes of evaluating mental health, benefiting both clients and healthcare facilities?

Target Customer: Any person who is suffering from any mental health condition within any age range would be the customer.

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Criteria:

- The types of mental illnesses that can be detected by this must have 5 or more unique symptoms for the final prediction to be based off of.
- Margin of error of the solution should not exceed 8%.
- The solution should not be age or gender specific.

Constraints:

- Has to collect data to detect mental health conditions.
- Should have security to avoid data leaks.
- Solutions need to be backed up by credible sources.
- Solutions must cost next to nothing for the user.
- The output data of the patient must be public to only the healthcare providers and patient.
- Solutions must make the entire process efficient thereby solving the problem.
- Technology of the solutions must be updated as needed.

Generate Concepts:

After many group discussions and brainstorming sessions, we came up with the following 3 ideas.

1. MHIG (Mental Health Information Gatherer)
2. Health and Fitness Tracking Watch
3. BCIs (Brain Computer Interfaces)

Decision Matrix:

Solution	Covers many Depression Types	Cost	Data Collection	User friendliness & efficiency	Security	Total
MHIG	**3**	**5**	**5**	**5**	**5**	**23**
Health & Fitness Tracking Watch	3	3	3	3	5	17
BCIs	5	1	5	1	5	17

Our conclusion was to continue to make a prototype MHIG since it had the highest rankings in all our requirements.

Develop a Solution:

First we came up with ideas on how we are going to approach this concept. We decided it would be best to make our solution a program since that would be very flexible and useful in many ways. Next, the team did weeks worth of research on different depressive disorders and compared it to how they are evaluated in regular doctor office visits and how that's not always reliable. Then, a final list of seven main depressive disorders was put together: major depression, persistent depression, bipolar disorder (hypomania/depressive extremes), seasonal depression, psychotic depression, & atypical depression. After all research was done on these disorders, a list of symptoms per disorder was derived. Then, after a heavy extent of critical thinking on human

behavior, questions were generated for each symptom. The team carefully made sure that the questions were generated while keeping into consideration human flaw, bias, and ego. Only after all background research was done, and questions were ready to be encoded into our prototype we started to code. Our first attempt was to try to make a prototype on code.org since no one in our group has had a heavy coding background.

Construct & Test and Evaluate Solution:

Prototype 1: Code.org

Our first prototype was made on code.org using simple JS and prebuilt block code functions. Though this prototype of MHIG did not achieve all the goals (including criteria and constraints) that we wanted to achieve, it helped us create a plan of how we are actually going to code the next attempt of our prototype. Through code.org, we were able to learn what our framework and structure was going to be, and what kind of math and algorithms we had to make. This first prototype helped set up the basic idea of how we are going to format our questions and how the program would flow.

Prototype 2: VSC

Our second prototype was made on Visual Studios Code, incorporating JS, HTML, and CSS. Now that we had decided what algorithms and math we had to do we could focus on bettering our run time in code and finding efficient loops to make our program better. Though no one in our group had great experience in raw coding, some people had a little bit of prior knowledge. Hence, we decided it would be best to learn and incorporate new methods learned in the languages since it fits our criteria and constraints better. By building our program on VSC, we were able to connect our prototype to a database to store and collect all our data. Though our group ran into multiple errors, after multiple testing tries we were able to get through it and made sure our program ran smoothly.

Before coding, we had to plan the path of how a user would sign up or how a returning user would login, since it was a new feature we wanted to incorporate. The visual plan for that is added in the Visual Representation section of the portfolio. Thanks to the more features we could take advantage of in VSC, we also had access to more design elements, making the UI of our program aesthetically pleasing to the eye, as well as incorporating colors that are soothing.

Conclusion:

After creating the initial prototype, visualizing a plan, and doing numerous test runs to confirm the flow of the program, we developed our final model of MHIG. This program meets all criteria and constraints that we needed and allowed for flexible uses of the program to be used in different environments and scenarios as well. MHIG was made in such a way that the healthcare system, along with patients, can use the program to their advantage and preferences. For example, not only could the hospital make mental health tests a part of regular checkups, but they could also upload MHIG to their official site, so if patients or family members of patients are worried, they could take the test to gather symptoms before visits. Along with giving relief to the patient from the comfort of their home, this could save lots of time on the doctors' end as well, as the program gives them prior knowledge on where possible issues could be in one's mental health, and how to fix them. Moreover, the database and tracking system of MHIG allows the healthcare professionals to record and store symptoms of a patient to allow further analysis of how the patient is developing per each visit. This helps psychologists and psychiatrists have a better understanding of the patient, and allows them to visualize and figure out what core issues may be.

Communication of Solution

The grand engineering challenge for this year, Advance Health Informatics, helps doctors personalize care for their patients. They do this by analyzing large collections of medical data and looking for patterns and trends which will assist them in treating the patient. Our app provides personalized feedback about patients' mental health by collecting some simple data from the user and answering multiple-choice questions.

Our solution - the Mental Health Information Gathering app (MHIG) - is precisely coded to give us feedback on our mental health. Our MHIG app collects data on age, and user responses to questions per with various depression disorders, which aligns with the challenge's goal of using data to personalize care, as their data can be analyzed by doctors. These objective questions are not suggestive, and provide a range of clear-written and understandable answer choices from extreme to minimal, which allows the patient to select how they are feeling with clarity. Unlike answering in a doctor's office, there is a "prefer not to say" option which protects the patients' privacy.

By factoring in age and existing conditions, our app can tailor its feedback and provide recommendations to individual users according to the results of our survey. Our app promotes self-care, it allows users to assess their health and try to follow the practices suggested before having to go to a therapist. It saves time for both parties; the doctors and their patients. When the user gets their feedback on the spot, they will be advised a plan that helps their mental health grow, or it will advise them to visit a doctor. This predicts health outcomes, as per the challenge, and prevents issues before they escalate into severe disorders.

This also reduces the reliance on human intervention. By providing automated, yet reliable and consistent self-assessment and feedback, our app reduces the need for constant doctor consultations or therapy sessions. This can also be helpful for individuals who might face obstacles to traditional mental health care due to cost, accessibility, or transportation for the services, as our online MHIG program can be accessed from home.

Plan of Work Log

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
1. 12/4/23 - 12/5/23	Research, brainstorm, and identify the problem	6 hours	A.V., A.D., L.P, T.V., S.C., G.S.V	Over the course of two meetings we brainstormed and created a list of different problems in healthcare, including ones that specifically relate to health informatics.
2. 12/13/23 & 12/16/23	Develop the criteria, constraints, and possible solutions	7 hours	A.V. & T.V.	Generating three possible solutions to the problem that follow the criteria and constraints: MHIG, Fitness & Health Care Watch, Brain Controlled Interfaces.
3. 12/21/23	Select our final solution	1 hour	A.V., A.D., L.P, T.V., S.C., G.S.V	Using a decision matrix (part of the design process), we selected the best possible solution to our problem: Mental Health Information Gatherer, also known as MHIG.
4. 1/2/24	Finalize the details of our solution	1 hour	S.C., A.D.	We decided to make a program that asks the user multiple-choice questions, which gathers and assesses symptoms of certain depressive disorders.
5. 1/16/24	Begin coding the program on code.org	1 hour	L.P.	Using simple JavaScript and the prebuilt block code function, we created a basic plan and framework of how the program will look and run.
6. 1/20/24	Creating the homescreen & question screen of the app on VSC	3 days	L.P.	I designed the spatial organization of text, images, and buttons, along with the colors, and coded it on Visual Studios Code using JS, HTML, CSS.

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	**Task**	**Time involved**	**Team member responsible** (student initials)	**Comments**
2/23/24 1.	Reviewing old solution	2 hours	A.V., A.D., L.P, T.V., S.C., G.S.V	Reviewed mistakes made on old solution, tried to find room for improvement
3/1/24 2.	Brainstorming ideas, assigning roles	4 hours	A.V., A.D., L.P, T.V., S.C., G.S.V	Established the basic idea, assigned roles in the team for the smooth functioning of the project
3/8/24 3.	Modifying main solution, coding new platform	3 hours	A.V., A.D., L.P, T.V., S.C., G.S.V	Revising the primary solution to perfectly align with the problem statement and goals. Started coding for the website to run
3/17/24 4.	Planning Question format, portfolio	4 hours	L.P, S.C, A.V	Creating a plan to make the app user friendly, easily accessible and brainstorming new solutions
3/23/24 5.	Making Questions, Portfolio, App development	4 hours	T.V, G.S.V, A.V	Finished with the list of questions, modifying portfolio pages, and fixing errors in the code
3/29/24 6.	Tri-fold, Portfolio, App development	6 hours	A.V., A.D., L.P, T.V., S.C., G.S.V	We came up with a new Tri-fold design. Consequently updated the portfolio. Fixed errors in the code.

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